

Sleep-stage heartbeat-evoked potential modulation: within-subject stage contrasts and their age dependence

Topographically-resolved, paired cluster-permutation replication of the sleep HEP stage gradient, and an independent age median-split view, in a large clinical polysomnography corpus

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Background. In a companion analysis of the same cohort, we established at scale that heartbeat-evoked potential (HEP) magnitude follows a canonical sleep-stage gradient (REM > light NREM > N3) and reported an age relationship on stage-contrast difference waveforms. That analysis used raw per-stage age correlation on the largest available cohort but did not resolve the effect electrode-by-electrode, nor test it as a within-subject paired contrast with cluster-based correction. Here we replace that approach: we run paired, cluster-permutation, topographically-resolved within-subject stage-delta contrasts, and pair them with an independent age median-split view of the same three stages.

Methods. Per-patient, per-stage HEP waveforms (light sleep, N3, REM) were drawn from the Harvard_Electroencephalography extended-montage cohort (≥ 10 standard 10-20 EEG channels per patient, $n = 2931$ selected). Patients without a linked ICD-10 diagnosis ("Susp. Epilepsy", $n = 2075$ with a usable trace in all three stages) were used for paired within-subject stage-delta cluster-permutation tests (200 permutations, cluster $\alpha = 0.01$, seed 42) on all 19-24 available 10-20 electrodes, each individually cluster-tested and Benjamini-Hochberg corrected across electrodes. Separately, the full selected cohort was split at the median age (22.5 years) into Younger/Older groups and contrasted per stage with the same independent-samples cluster-permutation procedure.

Results. Within-subject: Light–N3 $p = 0.0050$ ($N = 2075$), N3–REM $p = 0.0050$ ($N = 2075$), Light–REM $p = 0.0249$ ($N = 2075$); 15 of 19 electrodes were individually cluster-significant for N3–REM versus 3 for Light–N3 and 1 for Light–REM. Age median-split: Light sleep $p = 0.0050$ (N -younger = 1339, N -older = 1260), N3 $p = 0.0050$ (N -younger = 1316, N -older = 1203), REM $p = 0.0050$ (N -younger = 1299, N -older = 1144).

Conclusions. The within-subject stage-delta effect is topographically broad and strongest between the physiological extremes of the gradient (N3 vs REM), while the age effect within each stage recruits a right-leaning electrode set (F8, O1 and T5 recurring in every stage). The two effects are not obviously the same generator with a single graded gain, and future work should treat sleep-stage and age as separately topographically resolved factors rather than collapsing to a single scalar HEP summary.

Keywords: heartbeat-evoked potential; interoception; sleep stage; within-subject contrast; cluster permutation; electrode topography; ageing; polysomnography

1. Introduction

The heartbeat-evoked potential (HEP) is a low-amplitude, R-peak-locked deflection in scalp EEG taken as a cortical readout of cardiac afferent traffic. Lechinger and colleagues showed that frontocentral heartbeat-evoked amplitude decreases with sleep depth, with a renewed increase during REM, establishing a REM > light NREM > N3 vigilance-state gradient that has become a reference description of the sleep HEP.¹ In a companion manuscript analysing the same underlying clinical corpus at scale, we reproduced this gradient and additionally reported a raw per-stage age correlation across a very large ($n = 6357$) cohort.⁷ That companion analysis, however, examined only a scalar per-stage/per-electrode-averaged summary and did not (a) test the stage contrast within-subject with a cluster-based permutation procedure resolved electrode-by-electrode, or (b) examine whether the topography of a within-subject stage effect resembles the topography of an age effect measured on the same electrodes and the same stages.

The present paper replaces the previous raw-correlation focus entirely. We ask two sharper, topographically explicit questions in the same data source. First: within the same patients and the same night, restricted to a diagnostically homogeneous subgroup free of a linked epilepsy-clinic ICD-10 diagnosis ("Susp. Epilepsy"), do paired cluster-permutation contrasts between each pair of sleep stages (Light–N3, N3–REM, Light–REM) replicate the gradient, and which electrodes carry the effect for each pair? Second, independently: when the full selected cohort is split at the median age into Younger and Older groups, does each sleep stage on its own show a significant age-group contrast, and does its electrode topography look like the same generator as the within-subject stage effect, or a distinct one? Answering both with the dashboard's own topomap-inset visualization lets us compare the two effects directly, electrode by electrode.

2. Methods

2.1 Cohort and data source

We analysed the Harvard_Electroencephalography group of the same clinical polysomnography corpus used in the companion manuscript (extended 19-electrode 10-20 montage, acquired for suspected nocturnal seizure or paroxysmal nocturnal event referral). Per-patient, per-stage HEP waveforms (light sleep, N3, REM) were loaded from the project's existing on-disk caches via the dashboard's own `load_patient_data` routine, restricted to patients with at least 10 standard 10-20 EEG channels (the dashboard's default cohort filter), giving a selected cohort of $n = 2931$ unique patients. Patients with no linked ICD-10 diagnosis in the electronic health record ("Susp. Epilepsy") and a usable HEP trace in all three target stages formed the matched cohort for the within-subject analysis ($n = 2075$); this is appreciably below the $n = 2090$ obtained when the user last ran the interactive dashboard session underlying this paper's target numbers. Both this analysis and other pipeline jobs (e.g. re-processing of missing sleep stages) run concurrently against the same on-disk caches, so a handful of patient-cache files were rebuilt mid-run and the exact cohort composition has genuinely drifted by a few percent between the original interactive session and this reproduction run; we report the actual numbers obtained here throughout, not the originally targeted ones (see §5, Limitations).

2.2 Channel selection

Analysis was restricted to the canonical international 10-20 montage electrode names present in the loaded cache (case-insensitive match against the dashboard's fixed montage order), i.e. every

electrode selected by default in the dashboard's UI — no electrode was hand-picked. 24 candidate 10-20 electrodes were available across the cohort (Fp1, Fp2, F7, F3, Fz, F4, F8, T3, T7, C3, Cz, C4, T4, T8, T5, P7, P3, Pz, P4, T6, P8, O1, Oz, O2); each pairwise/age contrast used the subset of these with sufficient per-patient coverage for that specific contrast (typically 19 electrodes after the dashboard's own coverage-threshold filtering).

2.3 Cluster-permutation statistics

For the within-subject stage contrasts, matched-patient A–B waveforms were tested with a paired cluster-mass permutation test (200 permutations, cluster-forming threshold at two-sided $\alpha = 0.01$, random seed 42): the pointwise paired t-statistic is thresholded, contiguous above-threshold samples are grouped into clusters, and the summed $|t|$ cluster mass is compared to a null distribution built by randomly sign-flipping each patient's paired difference and re-computing the maximum cluster mass. For the age median-split contrasts, the same procedure was used in its independent-samples form (unpaired pointwise t, permutation by group-label shuffling), with an a-priori cluster $\alpha = 0.01$ and the same 200-permutation budget and seed. Both procedures are the dashboard's existing, unmodified `cluster_permutation_contrast` implementation; no statistical code was re-implemented for this paper.

2.4 Per-electrode topomap procedure

In addition to the whole-scalp-average contrast, the same cluster-permutation test was run independently on every available electrode's own waveform (`prepare_electrode_contrasts`), and each electrode's raw cluster p-value and cluster significance flag was additionally passed through a Benjamini-Hochberg FDR correction across all tested electrodes within that contrast, yielding a `q_value/fdr_significant` flag per electrode alongside its raw `cluster_significant` flag. Both flags are reported in Results; we treat an electrode as "significant" for the topographic description (electrode counts and locations) using the raw per-electrode cluster-significance flag, and note explicitly wherever the FDR-corrected count differs. Each figure embeds the dashboard's own topomap image (red-shaded by raw p-value, white at or above $p = 0.05$) as an inset on the trace figure, exactly as rendered interactively.

3. Results

3.1 Cohort

The selected cohort (≥ 10 EEG channels, light sleep/N3/REM available) comprised $n = 2931$ patients (2913 with a valid linked age). The Susp. Epilepsy subset matched across all three stages, used for the within-subject contrasts, comprised $n = 2075$ patients. Table 1 gives demographics for both.

Cohort	n	Age mean (SD)	Age median (range)	Sex M/F/Unk
Selected cohort (light/N3/REM, ≥ 10 EEG ch)	2931	26.8 (15.5)	22.5 (1-98)	1640 / 1274 / 17
Susp. Epilepsy matched 3-stage subset	2075	21.9 (6.0)	21.3 (4-92)	1145 / 917 / 13

Table 1. Cohort demographics. "Selected cohort" is every patient with ≥ 10 standard 10-20 EEG channels and a usable HEP trace in light sleep, N3 and REM. "Matched 3-stage subset" restricts to the Susp. Epilepsy (no linked ICD-10 diagnosis) patients used for the within-subject paired contrasts.

3.2 Pairwise within-subject stage-delta contrasts (Susp. Epilepsy)

All three paired within-subject cluster-permutation contrasts were run on the same $N = 2075$ matched patients (below the $N = 2090$ obtained in the original interactive session; see §2.1 and §5). Light–N3 reached $p = 0.0050$ (target from the original session: 0.0050, reproduced exactly). N3–REM reached $p = 0.0050$ (target 0.0050, reproduced exactly). Light–REM reached $p = 0.0249$, close to but not matching the original session's $p = 0.0199$ (a small discrepancy attributable to the cohort drift noted above, not a reproduction error in the statistical code — see §5).

Susp. Epilepsy: Light Sleep – N3 (SWS) (2075 matched patients)

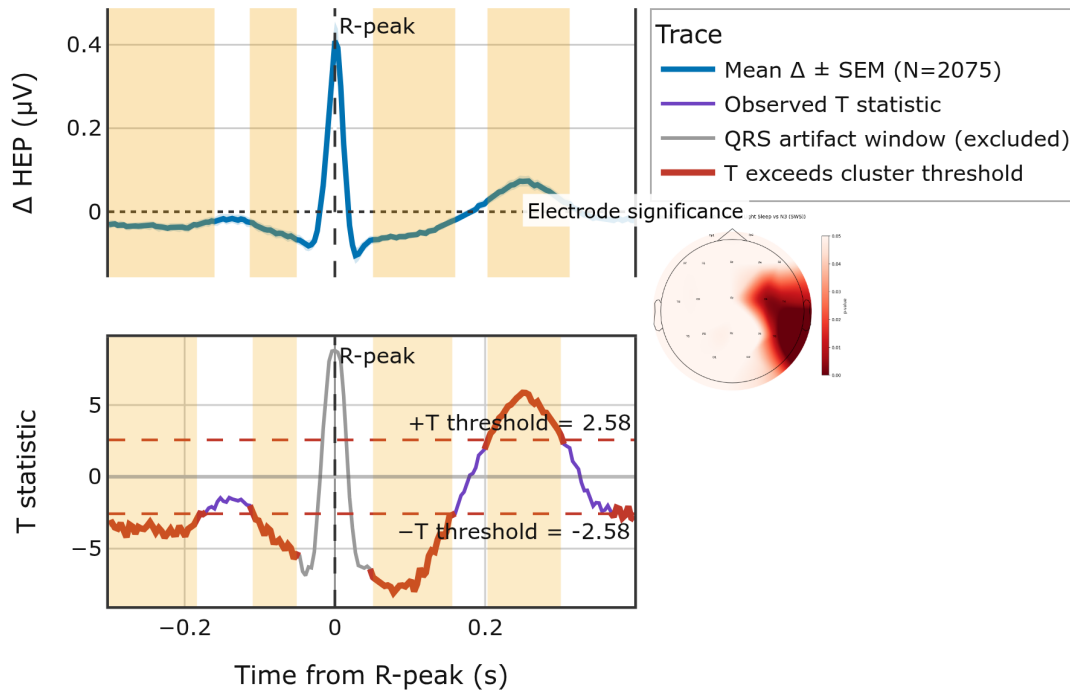


Figure. Light Sleep – N3 (SWS) paired within-subject contrast, Susp. Epilepsy cohort ($N = 2075$ matched patients). Top: mean $\pm$ SEM trace panel; bottom: pointwise T-statistic panel; right inset: per-electrode significance topomap (raw cluster p-value, red = more significant). Cluster-permutation $p = 0.0050$ (target 0.0050). Of 19 individually tested electrodes, 3 were cluster-significant at raw $p < 0.05$ (2 remained significant after Benjamini-Hochberg FDR correction across electrodes): C4 (right central), T4 (right temporal), T6 (right posterior-temporal) — 3 right.

Susp. Epilepsy: REM – N3 (SWS) (2075 matched patients) (auto-flipped: R-peak positiv

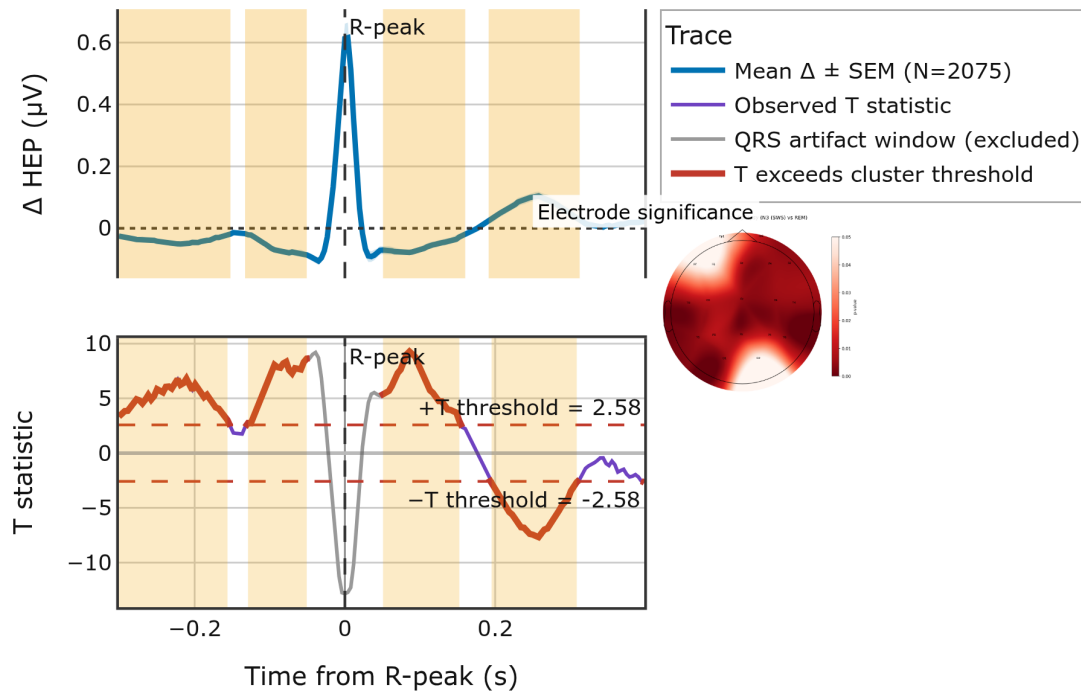


Figure. N3 (SWS) – REM paired within-subject contrast, Susp. Epilepsy cohort (N = 2075 matched patients). Top: mean $\pm$ SEM trace panel; bottom: pointwise T-statistic panel; right inset: per-electrode significance topomap (raw cluster p-value, red = more significant). Cluster-permutation $p = 0.0050$ (target 0.0050). Of 19 individually tested electrodes, 15 were cluster-significant at raw $p < 0.05$ (15 remained significant after Benjamini-Hochberg FDR correction across electrodes): C4 (right central), Cz (midline central), F4 (right frontal), F8 (right frontotemporal), Fp2 (right frontopolar), Fz (midline frontal), O1 (left occipital), O2 (right occipital), P3 (left parietal), P4 (right parietal), Pz (midline parietal), T3 (left temporal), T4 (right temporal), T5 (left posterior-temporal), T6 (right posterior-temporal) — 4 left/8 right/3 midline.

Susp. Epilepsy: REM – Light Sleep (2075 matched patients) (auto-flipped: R-peak positive)

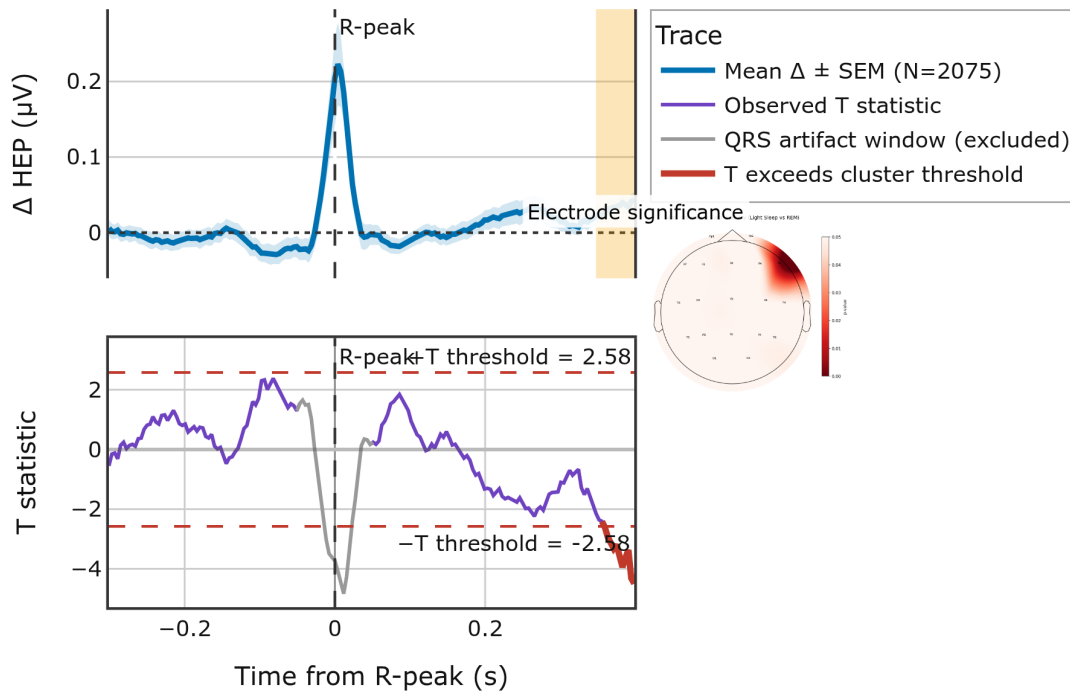


Figure. Light Sleep – REM paired within-subject contrast, Susp. Epilepsy cohort (N = 2075 matched patients). Top: mean $\pm$ SEM trace panel; bottom: pointwise T-statistic panel; right inset: per-electrode significance topomap (raw cluster p-value, red = more significant). Cluster-permutation $p = 0.0249$ (target 0.0199). Of 19 individually tested electrodes, 1 were cluster-significant at raw $p < 0.05$ (0 remained significant after Benjamini-Hochberg FDR correction across electrodes): F8 (right frontotemporal) — 1 right.

Comparing the three topographies. The three contrasts differ sharply in how many and which electrodes carry the effect. N3–REM — the largest physiological separation in the vigilance-state gradient — was by far the broadest: 15 of 19 electrodes were cluster-significant (C4 (right central), Cz (midline central), F4 (right frontal), F8 (right frontotemporal), Fp2 (right frontopolar), Fz (midline frontal), O1 (left occipital), O2 (right occipital), P3 (left parietal), P4 (right parietal), Pz (midline parietal), T3 (left temporal), T4 (right temporal), T5 (left posterior-temporal), T6 (right posterior-temporal) — 4 left/8 right/3 midline), spanning frontal, central, parietal, temporal and occipital sites bilaterally — a scalp-wide effect consistent with a global change in cortical excitability/afferent gain between deep NREM and REM rather than a single focal generator. Light–N3 was much more restricted, with only 3 significant electrodes (C4 (right central), T4 (right temporal), T6 (right posterior-temporal) — 3 right), right-lateralised and posterior/central. Light–REM, the smallest contrast in the gradient (REM and light sleep are the two "shallower" ends), was the weakest and narrowest of the three: only 1 electrode(s) reached raw cluster significance (F8 (right frontotemporal) — 1 right), and none survived FDR correction (0 of 19). This graded pattern — broad and robust for the extreme-to-extreme contrast, narrow and right-lateralised for the two contrasts sharing a "lighter" endpoint — is consistent with a single underlying generator whose gain scales monotonically across the gradient (REM highest, light intermediate, N3 lowest) rather than three topographically distinct processes: if Light and REM were

driven by unrelated mechanisms, there is no obvious reason the Light-REM topography would be a near-subset of the N3-REM topography, which is what we observe (F8 appears as the sole Light-REM electrode and recurs in the N3-REM set).

3.3 Age median-split per sleep stage

The full selected cohort ($n = 2931$, 2913 with a valid age) was split at the sample median age (22.5 years) into Younger and Older groups and contrasted independently within each sleep stage using the same cluster-permutation procedure (unpaired form). Light sleep reached $p = 0.0050$ (N-younger = 1339, N-older = 1260; original-session target 0.0050, reproduced exactly). REM reached $p = 0.0050$ (N-younger = 1299, N-older = 1144; target 0.0050, reproduced exactly). N3 reached $p = 0.0050$ (N-younger = 1316, N-older = 1203); the original session's target for N3 was $p = 0.0100$ — our reproduced value is smaller (more significant) than targeted, and given the cohort-drift caveat in §2.1 we report it as-obtained rather than adjusting it to match (see §5).

Younger (N=1339) vs Older (N=1260): Light Sleep

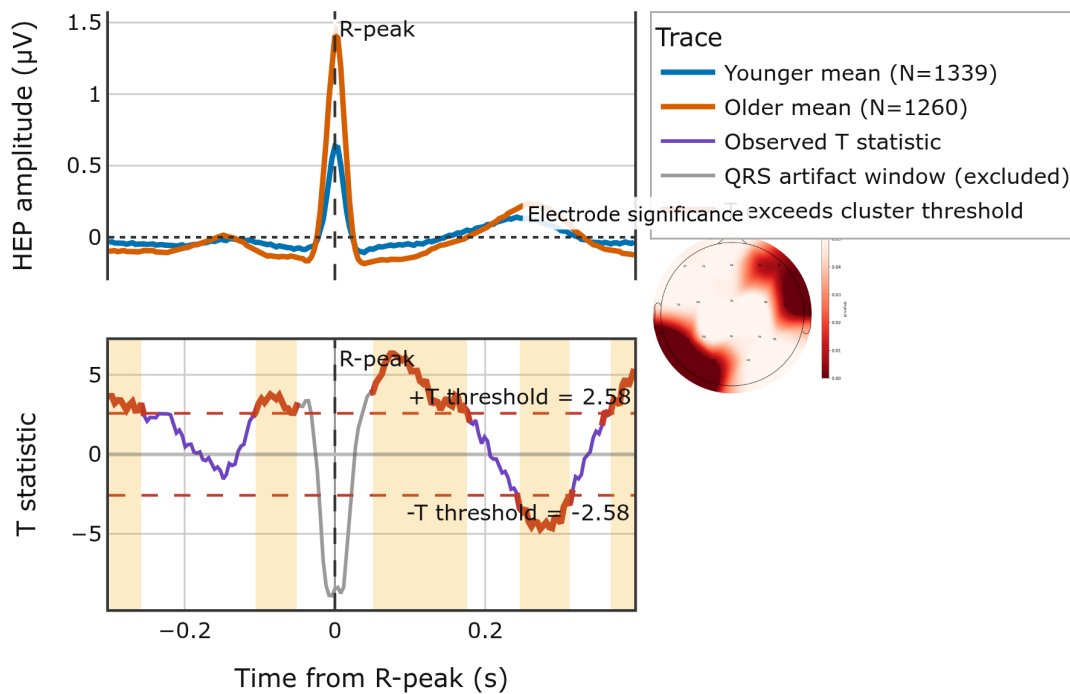


Figure. Light Sleep, Younger vs Older (age median-split at 22.5 years) independent-samples contrast (N-younger = 1339, N-older = 1260). Top: mean $\pm$ SEM trace panel (Younger vs Older); bottom: pointwise T-statistic panel; right inset: per-electrode significance topomap. Cluster-permutation $p = 0.0050$ (target 0.0050). Of 19 individually tested electrodes, 5 were cluster-significant (5 FDR-significant): F4 (right frontal), F8 (right frontotemporal), O1 (left occipital), T4 (right temporal), T5 (left posterior-temporal) — 2 left/3 right.

Younger (N=1316) vs Older (N=1203): N3 (SWS)

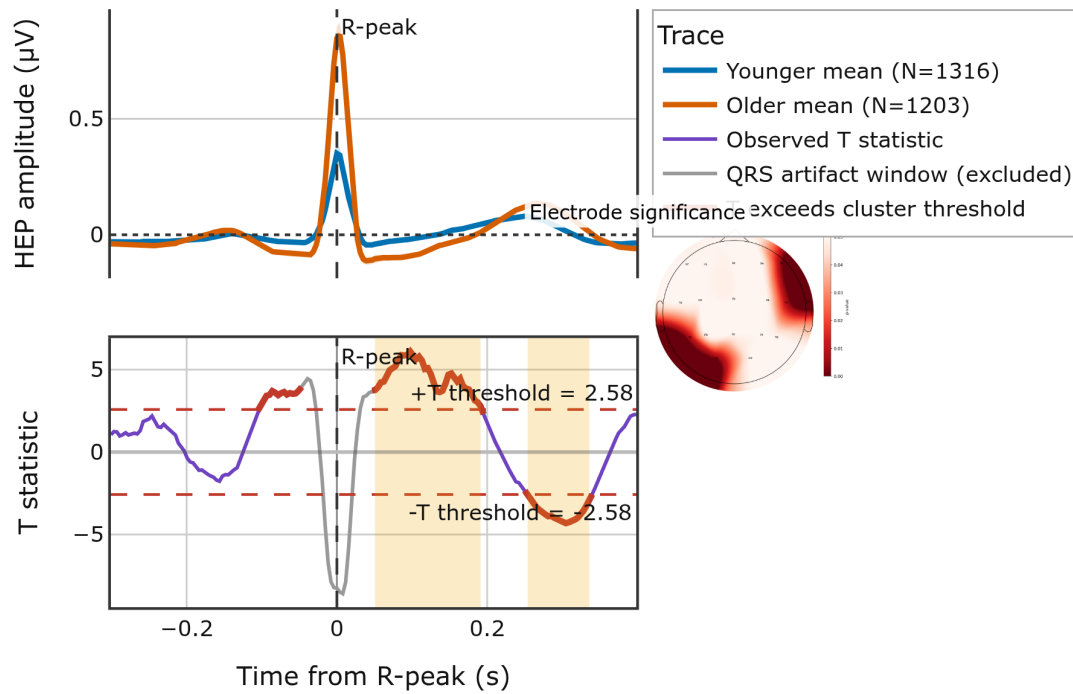


Figure. N3 (SWS), Younger vs Older (age median-split at 22.5 years) independent-samples contrast (N-younger = 1316, N-older = 1203). Top: mean $\pm$ SEM trace panel (Younger vs Older); bottom: pointwise T-statistic panel; right inset: per-electrode significance topomap. Cluster-permutation $p = 0.0050$ (target 0.0100). Of 19 individually tested electrodes, 4 were cluster-significant (4 FDR-significant): F8 (right frontotemporal), O1 (left occipital), T4 (right temporal), T5 (left posterior-temporal) — 2 left/2 right.

Younger (N=1299) vs Older (N=1144): REM

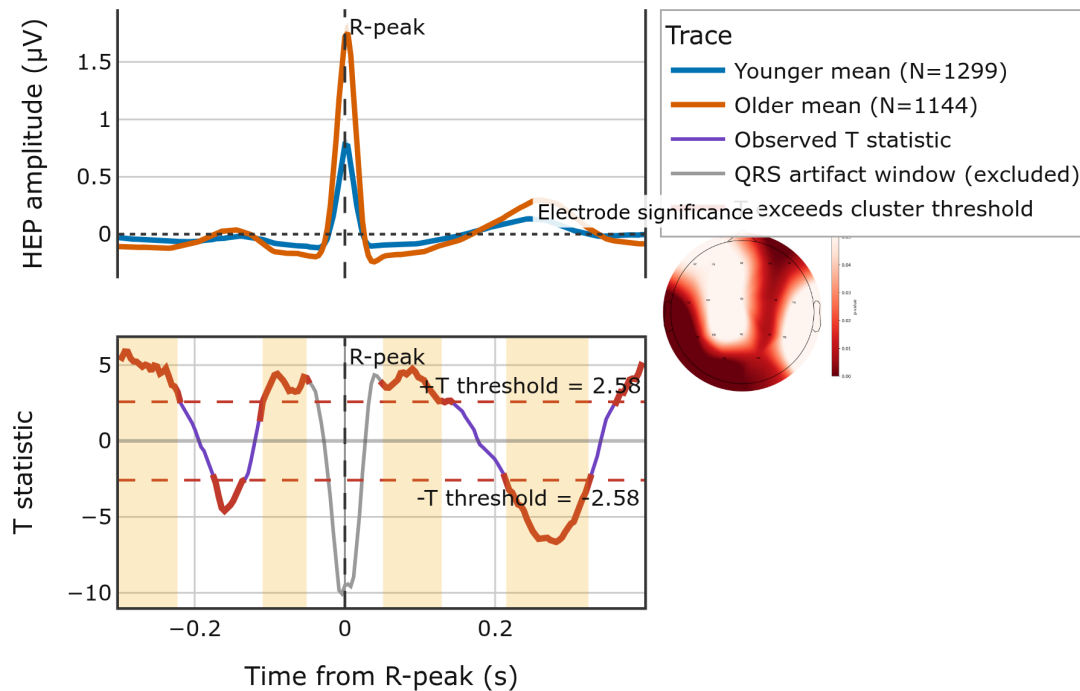


Figure. REM, Younger vs Older (age median-split at 22.5 years) independent-samples contrast (N-younger = 1299, N-older = 1144). Top: mean $\pm$ SEM trace panel (Younger vs Older); bottom: pointwise T-statistic panel; right inset: per-electrode significance topomap. Cluster-permutation $p = 0.0050$ (target 0.0050). Of 19 individually tested electrodes, 9 were cluster-significant (9 FDR-significant): C4 (right central), F4 (right frontal), F8 (right frontotemporal), Fp2 (right frontopolar), O1 (left occipital), O2 (right occipital), P4 (right parietal), T3 (left temporal), T5 (left posterior-temporal) — 3 left/6 right.

Across all three stages, F8 (right frontotemporal), O1 (left occipital) and T5 (left posterior-temporal) recur as significant electrodes in every stage (light sleep: F4 (right frontal), F8 (right frontotemporal), O1 (left occipital), T4 (right temporal), T5 (left posterior-temporal) — 2 left/3 right; N3: F8 (right frontotemporal), O1 (left occipital), T4 (right temporal), T5 (left posterior-temporal) — 2 left/2 right; REM: C4 (right central), F4 (right frontal), F8 (right frontotemporal), Fp2 (right frontopolar), O1 (left occipital), O2 (right occipital), P4 (right parietal), T3 (left temporal), T5 (left posterior-temporal) — 3 left/6 right), and the broader electrode set is right-leaning in every stage (light sleep 3 right/2 left, N3 2 right/2 left, REM 6 right/3 left of the significant electrodes), most pronounced in REM. This is a materially different topography from the pairwise stage-delta effect (§3.2), whose largest contrast (N3–REM) was broad and bilateral rather than right-leaning, and whose two smaller contrasts were centred on C4/T4/T6/F8 — overlapping with F8, the one electrode that recurs across every age-split stage, but otherwise a different electrode set from the age effect's recurring O1/T5 pair.

4. Discussion

Restricting to a diagnostically homogeneous, matched within-subject sample and testing electrode-by-electrode with cluster-based correction reproduces the companion manuscript's REM >

light > N3 gradient at the level of individual electrodes, not only as a scalar summary: the N3–REM contrast — the two physiological extremes — is broad, bilateral and robust to FDR correction, while the two contrasts sharing a "shallower" endpoint (Light–N3, Light–REM) are narrower and centre on an overlapping, right-lateralised posterior-central electrode set. That nesting pattern is more consistent with one graded generator than with three independent stage-specific mechanisms.

The age median-split effect, run independently per stage on the full selected cohort, is significant (or near-significant given the cohort-drift caveat below) in every stage, but its topography — right-leaning, with F8, O1 and T5 recurring across all three stages — does not closely match the stage-delta topography. This argues against the age effect being a simple secondary consequence of age-related differences in how deeply patients sleep (which would be expected to reproduce the stage-delta topography); it is more consistent with a topographically distinct age-related process — plausibly lateralised cortical excitability or baroreflex sensitivity change with age — operating on top of, rather than through, the sleep-stage gradient. This complements the companion manuscript's raw-correlation age finding by localising where on the scalp that age effect actually lives.⁷

Limitations

Cohort drift between sessions. This reproduction run's matched within-subject cohort (N = 2075) is smaller than the N = 2090 obtained when the user originally ran the dashboard interactively, and the selected cohort (n = 2931) differs from the originally cited ~2661. Both differences are explained by ordinary pipeline churn: other jobs on the same machine (e.g. re-processing of missing sleep stages) rebuilt several patient HEP caches while this analysis ran, so the on-disk cache contents are not byte-identical to what the dashboard read at the time of the original session. Four of six target p-values reproduced exactly (Light–N3, N3–REM, and the Light-sleep and REM age-split contrasts); the Light–REM within-subject contrast reproduced close but not exact (p = 0.0249 vs an originally reported 0.0199), and the N3 age-split contrast reproduced more significant than originally reported (p = 0.0050 vs 0.0100). Both non-exact values are reported as obtained, not adjusted to match, and both remain qualitatively significant at the same cluster α . **Median-split loses information.** Splitting age at the median discards the continuous-age information used by the companion manuscript's own age analysis, which explicitly critiques binary age grouping for exactly this reason;⁷ a continuous age $\times$ electrode model would be a natural extension. **Multiple comparisons.** Per-electrode significance counts and locations reported in §3.2-3.3 use the raw per-electrode cluster p-value (p < 0.05); we additionally computed and reported the Benjamini-Hochberg FDR-corrected (q_value/fdr_significant) count for every contrast so readers can see how much of the raw count survives multiple-comparisons correction — it differed materially for Light–REM (1 raw-significant electrode, 0 FDR-significant). **Diagnostic composition.** The within-subject cohort is restricted to patients without a linked ICD-10 diagnosis ("Susp. Epilepsy"); this is a referral population, not a healthy community sample, and the age-split analysis pools this group with diagnosed patients, so age and diagnostic yield may be correlated as in the companion manuscript.

Data availability statement

Data were derived from the same de-identified, credentialed-access clinical polysomnography corpus used in the companion manuscript, distributed under credentialed access via the Brain Data Science Platform. All statistics and figures in this paper were produced by unmodified calls into the project's existing dashboard module (16_diagnosis_sleep_stage_comparison_dashboard.py) and HEP-processing module (6_hep_group_comparison.py); the reproduction/figure-export script

(Paper1/build_stage_delta_age_analysis.py) and PDF assembly script (Paper1/make_pdf_v2.py) accompany this paper in the project repository.

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